

Pion EMFF Background-Response Diagnostic

Finite-Field Energy Shift versus Linearized Response Propagator

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1 Purpose

This note documents the pion electromagnetic form-factor (EMFF) background-response diagnostic implemented in

`pyquda_measurement_utils/pion_current_background_response_vibe_develop.py`

and

`application/EMFF_pion_background_response/perlmutter/.`

The goal is to study a background-field/Feynman-Hellmann route to pion EMFF matrix elements while *not* modifying the QUDA Dirac operator and while *not* performing inversions with a finite external field $D + \lambda\mathcal{O}$. Instead, the current implementation constructs the first-order response propagator using ordinary D^{-1} inversions.

This is not yet the full finite- λ background-field method. It is the linearized method at $\lambda = 0$, which is mathematically equivalent to a summed current insertion.

The same source module now also contains a minimal second-order current-current response building block. This is intended for future hadronic-tensor or Compton-amplitude style diagnostics, while the present production application remains the first-order pion EMFF response.

2 Two Related But Distinct Methods

2.1 Finite- λ background-field method

The most literal background-field method modifies the action or Dirac operator,

$$D_\lambda = D + \lambda\mathcal{O}, \quad (1)$$

then computes ordinary two-point functions in this modified theory,

$$C_\lambda(t) = \langle O_\pi(t) O_\pi^\dagger(0) \rangle_\lambda. \quad (2)$$

At large Euclidean time,

$$C_\lambda(t) = Z_\lambda e^{-E_\lambda t} + \dots. \quad (3)$$

The desired matrix element is obtained from the energy response,

$$\left. \frac{\partial E_\lambda}{\partial \lambda} \right|_{\lambda=0}. \quad (4)$$

This is the usual “extract the energy shift” picture of the background-field method.

2.2 Linearized response at $\lambda = 0$

The method currently implemented in this repository does not invert $D + \lambda\mathcal{O}$. Instead, it expands the propagator,

$$S_\lambda = (D + \lambda\mathcal{O})^{-1} = S - \lambda S\mathcal{O}S + O(\lambda^2). \quad (5)$$

Thus,

$$\left. \frac{\partial S_\lambda}{\partial \lambda} \right|_{\lambda=0} = -S\mathcal{O}S. \quad (6)$$

The code constructs the object

$$S_{\text{resp}} = S\mathcal{O}S = D^{-1}\mathcal{O}S, \quad (7)$$

using ordinary inversions. Therefore, relative to the derivative in Eq. (6), the physical finite-difference sign is

$$\left. \frac{\partial S_\lambda}{\partial \lambda} \right|_{\lambda=0} = -S_{\text{resp}}. \quad (8)$$

The positive-sign S_{resp} convention is useful because it can be compared directly against the explicit summed three-point function.

2.3 Why the linear response looks like a summed three-point function

Expanding the two-point function in the external source gives

$$\left. \frac{\partial C_\lambda(t)}{\partial \lambda} \right|_{\lambda=0} = - \sum_{\tau} \langle O_\pi(t) \mathcal{O}(\tau) O_\pi^\dagger(0) \rangle, \quad (9)$$

up to the sign convention chosen in the action perturbation. In the code-sign convention of Eq. (7), the response contraction is compared to

$$\sum_{\tau} C_3(t, \tau). \quad (10)$$

This explains why a linearized background-response calculation has analysis features similar to a summed three-point function: it is the derivative $\partial C_\lambda / \partial \lambda|_0$, not a finite- λ energy fit.

3 Pion EMFF Kinematics

The pion EMFF matrix element is defined by

$$\langle \pi(p_f) | J_\mu(0) | \pi(p_i) \rangle = (p_i + p_f)_\mu F_\pi(Q^2), \quad q = p_f - p_i. \quad (11)$$

The lattice implementation uses

$$\mathbf{p}_i = \mathbf{p}_f - \mathbf{q}_{\text{ext}}. \quad (12)$$

The explicit three-point correlator is

$$C_3^\Gamma(\mathbf{q}, \tau; \mathbf{p}_f, t_{\text{sep}}) = \sum_{\mathbf{x}} e^{-i\mathbf{q} \cdot (\mathbf{x} - \mathbf{x}_0)} \text{tr} [S_{\text{seq,anti}}(\mathbf{x}, \tau; \mathbf{p}_f, t_{\text{sep}}) \Gamma S_q(\mathbf{x}, \tau; \mathbf{x}_0, 0) \Gamma_{\text{src}}], \quad (13)$$

where the fixed-sink sequential propagator has already absorbed the sink momentum $\mathbf{p}_f$, the sink time t_{sep} , and the sink interpolator Γ_{sink} . The standard EMFF current is the temporal vector current,

$$\Gamma = \gamma_T, \quad (14)$$

labeled T in the code.

4 Current-Inserted Source

For a local current with momentum transfer $\mathbf{q}$,

$$\mathcal{O}_{\Gamma, \mathbf{q}}(x) = \Gamma \Phi_{\mathbf{q}}(x - \mathbf{x}_0), \quad (15)$$

where $\Phi_{\mathbf{q}}$ is produced by `MomentumPhase` using the same phase convention as the ordinary EMFF code.

Given an ordinary source-to-current propagator

$$S(x, x_0), \quad (16)$$

the all-time current-inserted source is

$$\eta_{\Gamma, \mathbf{q}}(x) = \Gamma \Phi_{\mathbf{q}}(x - \mathbf{x}_0) S(x, x_0). \quad (17)$$

The response propagator is then obtained by a standard inversion,

$$S_{\text{resp}}(y, x_0; \Gamma, \mathbf{q}) = \sum_x S(y, x) \Gamma \Phi_{\mathbf{q}}(x - \mathbf{x}_0) S(x, x_0). \quad (18)$$

In code, Eq. (17) is built by

```
build_local_current_inserted_source(...)
```

and Eq. (18) by

```
invert_local_current_response_propagator(...).
```

5 Response Correlator

The response propagator can be contracted like the forward quark line in a pion two-point function. With an ordinary antiquark line S_{anti} ,

$$C_{\text{resp}}(t; \mathbf{p}_f, \mathbf{q}) = \sum_{\mathbf{y}} e^{-i\mathbf{p}_f \cdot (\mathbf{y} - \mathbf{x}_0)} \text{tr} [S_{\text{anti}}(\mathbf{y}, t; \mathbf{x}_0, 0) \Gamma_{\text{sink}} S_{\text{resp}}(\mathbf{y}, t; \mathbf{x}_0, 0; \Gamma, \mathbf{q}) \Gamma_{\text{src}}]. \quad (19)$$

For the code-sign convention,

$$C_{\text{resp}}(t_{\text{sep}}; \mathbf{p}_f, \mathbf{q}) = \sum_{\tau} C_3^{\Gamma}(\mathbf{q}, \tau; \mathbf{p}_f, t_{\text{sep}}), \quad (20)$$

provided the same current gamma, source/sink interpolators, momentum phases, and smearing conventions are used.

The first validated case is

$$\mathbf{p}_f = \mathbf{0}, \quad \mathbf{q} = \mathbf{0}, \quad \mathbf{p}_i = \mathbf{0}, \quad \Gamma = \gamma_T, \quad \Gamma_{\text{src}} = \Gamma_{\text{sink}} = \gamma_5. \quad (21)$$

On the S8T32 test gauge, the diagnostic found

$$\sum_{\tau} C_3(\tau) = 327.27206385480486 + 0.1342484007638715 i, \quad (22)$$

$$C_{\text{resp}} = 327.27206385480486 + 0.13424840076386432 i, \quad (23)$$

with relative difference

$$2.2 \times 10^{-17}. \quad (24)$$

6 Energy-Shift Interpretation

Although Eq. (20) looks like a summed three-point function, it is still directly related to the background-field energy shift. Suppose

$$C_{\lambda}(t) = Z_{\lambda} e^{-E_{\lambda} t}. \quad (25)$$

Expanding around $\lambda = 0$,

$$Z_{\lambda} = Z_0 + \lambda Z'_0 + O(\lambda^2), \quad (26)$$

$$E_{\lambda} = E_0 + \lambda E'_0 + O(\lambda^2), \quad (27)$$

gives

$$C_{\lambda}(t) = C_0(t) \left[1 + \lambda \left(\frac{Z'_0}{Z_0} - t E'_0 \right) \right] + O(\lambda^2). \quad (28)$$

Therefore

$$\frac{1}{C_0(t)} \left. \frac{\partial C_{\lambda}(t)}{\partial \lambda} \right|_{\lambda=0} = \frac{Z'_0}{Z_0} - t E'_0. \quad (29)$$

Thus, in the linear-response method, the energy-shift derivative is extracted from the slope of a response ratio, not from separate finite- λ energy fits. This is the same Feynman-Hellmann information, but represented at $\lambda = 0$.

7 Summed-Ratio Object

Define

$$R_{\text{sum}}(t_{\text{sep}}) = \frac{\sum_{\tau} C_3(t_{\text{sep}}, \tau)}{C_2(t_{\text{sep}})} = \frac{C_{\text{resp}}(t_{\text{sep}})}{C_2(t_{\text{sep}})}. \quad (30)$$

At large t_{sep} ,

$$R_{\text{sum}}(t_{\text{sep}}) = A - t_{\text{sep}} E'_0 + \text{excited-state corrections}, \quad (31)$$

where A absorbs overlap-factor response and contact-term effects.

For the finite- λ sign convention $D_{\lambda} = D + \lambda \mathcal{O}$, the derivative of the propagator has the sign in Eq. (6). The diagnostic code stores both

$$\text{response_sign} = +1$$

for direct comparison to the explicit summed three-point function, and

$$\text{finite_difference_derivative_sign} = -1$$

for the corresponding finite-field derivative convention.

8 Restricted Insertion-Time Windows

The full background-field energy-shift picture corresponds to an all-time insertion. However, in the linearized response implementation one may restrict the insertion time,

$$S_{\text{resp}}^{W_{\tau}} = \sum_{\tau \in W_{\tau}} D^{-1} [\delta_{t,\tau} \Gamma \Phi_{\mathbf{q}}(x - \mathbf{x}_0) S(x, x_0)]. \quad (32)$$

This produces

$$C_{\text{resp}}^{W_{\tau}}(t_{\text{sep}}) = \sum_{\tau \in W_{\tau}} C_3(t_{\text{sep}}, \tau). \quad (33)$$

Useful choices include

$$W_{\tau} = \text{all time slices}, \quad (34)$$

$$W_{\tau} = \{0, \dots, t_{\text{sep}}\}, \quad (35)$$

$$W_{\tau} = \{1, \dots, t_{\text{sep}} - 1\}, \quad (36)$$

$$W_{\tau} = \{\tau_{\text{min}}, \dots, t_{\text{sep}} - \tau_{\text{min}}\}. \quad (37)$$

The restricted windows are not the pure finite-field energy-shift method. They are analysis tools for the linearized method, useful for:

- removing source/sink contact terms,
- comparing with explicit three-point functions in the same time window,
- studying excited-state contamination,
- checking the stability of summed-ratio slopes.

9 Current Implementation

9.1 Source helper

The helper

```
build_local_current_inserted_source(prop_forward, phase_q, current_gamma, tau)
```

constructs

$$\eta(x) = \Gamma_{\text{cur}} \Phi_{\mathbf{q}}(x - \mathbf{x}_0) S(x, x_0). \quad (38)$$

If `tau` is provided, the source is masked to that global insertion time. If `tau` is omitted, all time slices are included.

9.2 Response inversion

The helper

```
invert_local_current_response_propagator(...)
```

performs the ordinary D^{-1} inversion of the current-inserted source. If a list of times `tau_list` is given, the code first builds the summed time-window source and then performs one inversion. This deliberately avoids storing per- τ response propagators.

9.3 Response contraction

The helper

```
contract_response_pion_2pt(...)
```

contracts the response propagator with the ordinary antiquark line using the same pion two-point contraction infrastructure as the rest of the pion code.

9.4 Current-current response

The same helper module also contains a minimal nested current-current response prototype,

```
invert_current_current_response_propagator(...).
```

For two local currents O_1 and O_2 , it constructs

$$S_{\text{resp}}^{(2)} = D^{-1} O_2 D^{-1} O_1 S. \quad (39)$$

With

$$O_i(x) = \Gamma_i \Phi_{\mathbf{q}_i}(x - \mathbf{x}_0), \quad (40)$$

this corresponds to the ordered two-current insertion

$$S_{21}^{(2)} = S O_2 S O_1 S, \quad (41)$$

up to the sign convention chosen for finite- λ derivatives. The associated pion two-point-like response contraction is

$$C_{\text{resp}}^{(2)}(t_{\text{sep}}) = \sum_{\tau_2 \in W_2} \sum_{\tau_1 \in W_1} C_{JJ}(t_{\text{sep}}, \tau_2, \tau_1), \quad (42)$$

where C_{JJ} is the pion correlator with two local current insertions on the active quark line. The total momentum injected by the two currents is

$$\mathbf{q}_{\text{tot}} = \mathbf{q}_1 + \mathbf{q}_2, \quad \mathbf{p}_i = \mathbf{p}_f - \mathbf{q}_{\text{tot}}. \quad (43)$$

This is the direct local-current analogue of a two-current insertion into the quark line. It is intended as a diagnostic building block for current-current and hadronic-tensor style measurements. As with the first-order response, tau windows are summed at the inserted-source level before inversion; the code does not save per- τ response-propagator caches.

9.5 Diagnostic application

The Perlmutter diagnostic application is

`application/EMFF_pion_background_response/perlmutter/`.

The default run uses the S8T32 test gauge and compares

$$C_{\text{resp}}(t_{\text{sep}}) \quad \text{against} \quad \sum_{\tau} C_3(t_{\text{sep}}, \tau) \quad (44)$$

for $t_{\text{sep}} = 2$, $\Gamma = \gamma_T$, and zero external momentum. The upgraded diagnostic can also run several t_{sep} , several momentum transfers, and several current gamma matrices in a single job. The HDF5 output stores one record for each $(\Gamma, \mathbf{q}, t_{\text{sep}}, W_{\tau})$.

10 Implemented Upgrade Path

1. **Nonzero- q validation.** The response schema stores

$$\mathbf{p}_i = \mathbf{p}_f - \mathbf{q},$$

for every record. The phase test covers both ordinary and Breit-frame examples, including

$$\mathbf{p}_f = (0, 0, 1), \quad \mathbf{q} = (0, 0, 1), \quad \mathbf{p}_i = (0, 0, 0).$$

and

$$\mathbf{p}_f = (0, 0, 1), \quad \mathbf{q} = (0, 0, 2), \quad \mathbf{p}_i = (0, 0, -1).$$

2. **Multi- t_{sep} , C_2 , and summed-ratio output.** The application accepts a list of source-sink separations and saves

$$C_2(t_{\text{sep}}), \quad C_{\text{resp}}(t_{\text{sep}}), \quad R_{\text{sum}}(t_{\text{sep}}).$$

The saved ratios are

$$R_{\text{sum}}^{\text{resp}}(t_{\text{sep}}) = \frac{C_{\text{resp}}(t_{\text{sep}})}{C_2(t_{\text{sep}})}, \quad R_{\text{sum}}^{\text{explicit}}(t_{\text{sep}}) = \frac{\sum_{\tau \in W_{\tau}} C_3(t_{\text{sep}}, \tau)}{C_2(t_{\text{sep}})}. \quad (45)$$

3. **Restricted tau windows.** The supported choices are all time slices, source-to-sink, open source-to-sink, a symmetric restricted window, and an explicit integer range:

$$W_{\tau} = \{\tau_{\min}, \dots, t_{\text{sep}} - \tau_{\min}\}.$$

For restricted windows, the inserted source is summed before inversion.

4. **Multi-current gamma response suite.** The application accepts multiple gamma labels from the standard pion gamma list. The primary EMFF current is still V_4 , labeled T, but scalar, pseudoscalar, vector, axial, and tensor bilinears can be used as response diagnostics without changing the production EMFF contraction.
5. **HDF5 schema upgrade.** Version-2 output stores file-level kinematics and one `results/record_XXXX` group per diagnostic. Each record includes $\mathbf{p}_f$, $\mathbf{q}$, inferred $\mathbf{p}_i$, t_{sep} , current gamma, source/sink gamma labels, the tau-window definition, C_2 , explicit summed C_3 , response C_2 -like data, ratios, and the response-explicit difference. For analysis convenience, the same per-record scalars are also collected in `summary/` datasets:

$$\text{relative_difference}, \quad \text{response_R_sum}, \quad \text{explicit_R_sum},$$

together with $\mathbf{p}_f$, $\mathbf{q}$, $\mathbf{p}_i$, t_{sep} , gamma labels, and window labels. Analysis scripts can therefore scan the diagnostics without opening every `results/record_XXXX` group.

6. **Optional GPU regression tests.** The Perlmutter application remains a small S8T32 smoke workflow. The test suite contains CPU-only schema, phase, and algebra checks, while the GPU smoke run can be launched when a QUDA-enabled node is available.

Explicit non-goal. This implementation does not save a per- τ response-propagator cache. If arbitrary post-hoc tau windows become necessary, that should be considered as a separate design discussion rather than silently added to this diagnostic.

11 Summary

The present pion EMFF background-response code is a linearized Feynman-Hellmann diagnostic:

$$S_{\text{resp}} = D^{-1} \Gamma \Phi_{\mathbf{q}} S. \quad (46)$$

It does not compute finite- λ energies. Instead, it computes $\partial C_{\lambda} / \partial \lambda|_{\lambda=0}$, which equals a summed three-point function up to the convention-dependent sign. The energy-shift derivative is then encoded in the slope of the response ratio C_{resp}/C_2 as a function of t_{sep} .